

Breast Cancer Detection — CNN Classifier

A convolutional neural network for classifying breast histopathology patches as benign or malignant, optimized for recall to minimize missed diagnoses and support faster, more consistent specimen review.

PATHOLOGISTS

ONCOLOGY TEAMS

HEALTHCARE ORGANIZATIONS

AI-ASSISTED DIAGNOSTICS

EXPORT CASE REPORT

RECALL (MALIGNANT CLASS)

97.2%

PRECISION

84.1%

AUC

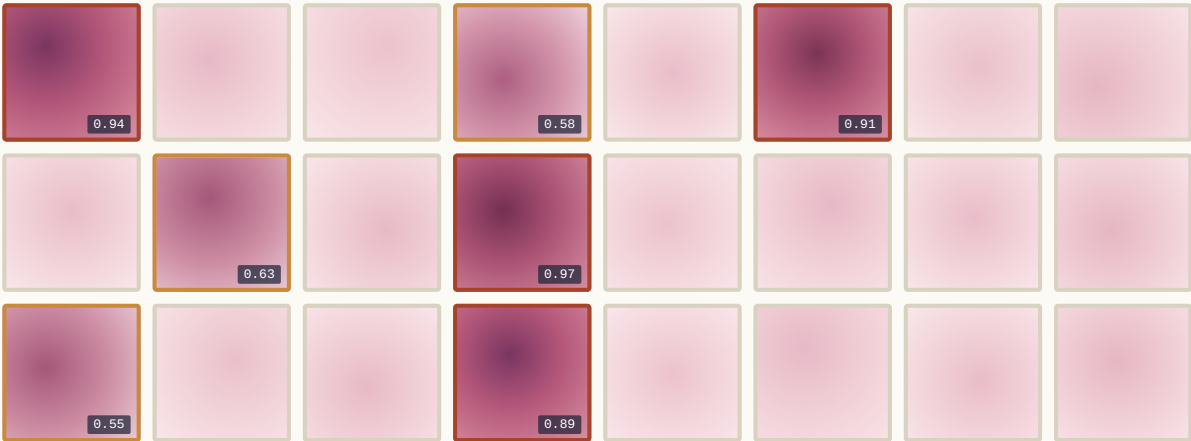
0.95

PATCHES FLAGGED TODAY

142 of 1,860

SPECIMEN PATCH REVIEW

224x224px · IDC histopathology set



☒ High-confidence malignant flag

☒ Borderline — recommend review

☐ Benign / clear

MODEL PERFORMANCE

held-out test set

	PRED. BENIGN	PRED. MALIGNANT
ACTUAL BENIGN	91.4%	8.6%
ACTUAL MALIGNANT	2.8%	97.2%

2.8% false-negative rate on malignant cases — the metric this model is tuned to minimize, given the cost of a missed diagnosis.

Recall (Sensitivity) — primary metric 97.2%



Precision 84.1%



F1 Score 90.2%



REVIEW QUEUE

sorted by confidence

CASE	CONFIDENCE	CLASS	STATUS
PT-04471		Malignant	Pending
PT-04466		Malignant	Pending
PT-04459		Malignant	In Review
PT-04452		Malignant	Pending
PT-04447		Malignant	Pending
PT-04440		Malignant	Pending

FILTERS

CONFIDENCE THRESHOLD

≥ 0.50 (Recall-Optimized)

SPECIMEN BATCH

Batch 2026-07-16

SORT QUEUE BY

Confidence (High → Low)

SO WHAT

The model is tuned to over-flag rather than miss — pathologists still make every diagnosis, but the queue surfaces likely-malignant patches first, cutting time-to-review on the cases that matter most.

01 — Diagnostic Support

Triage, not autonomous diagnosis

Flags high-risk patches for prioritized review — a clinician makes every call, not the model.

02 — Recall-Optimized Design

Tuned for the cost asymmetry in oncology

A missed malignant case costs far more than a false alarm, so the model is biased toward sensitivity.

03 — Workflow Integration

Scales with specimen volume

Supports faster, more consistent review across high-volume caseloads at scale.